

Questions (Answers)

① Database is generally a collection of data stored and accessed electronically from a computer system.

- Databases are essential in the modern world because they help organisations and businesses keep track of heaps of information, from customer details to transactions and even something as complex as data generated by IoT devices. They are like the backbone of most apps and systems we use daily.

- Real examples:

- * Facebook use databases to store user info, posts etc
- * Banks use them to manage transactions, accounts and customer data.
- * Online stores like Amazon rely on databases for inventory, orders, and customer management.

② Data warehouse is like a massive storage room for historical data, specifically designed for analysis and reporting.

Key differences:

- Purpose: databases are for transactions (CRUD - create, read, update, delete) while data warehouses are for analysis and reporting.
- Structure: databases are often normalized for efficiency, while data warehouses use schemas optimized for querying.
- Data volume: data warehouses handle massive amounts of historical data.
- * Companies use data warehouses to:
 - Analyse trends and patterns over time.

- ③ A data lake is like a massive reservoir that stores raw, unprocessed data in its original form. It's a centralized repo where you can dump structured, semi-structured and unstructured data, and then process it later for various uses.

Key differences betw data warehouse:

- * Data structure: Data lakes store data in its own form, while data warehouses store structured, processed data.
- * Schema: Data lakes use a "schema-on-read" approach (structure is applied when data is read), while data warehouses use "schema-on-write" (structure is applied when data is written).

Business prefer data lakes when:

- * They have large volumes of diverse data (like IoT, social media).
- * They want flexibility in how they process and analyze data.
- * They're exploring data science, machine learning or big data.

- ④ A Data Mart is like a mini-data warehouse, focused on a specific business area or department.

Relation to data warehouse:

- * A data mart is usually derived from a data warehouse, pulling in relevant data for a specific purpose.

Businesses prefer data warehouse when:

- * They need centralized, integrated data across the organization.
- * They want a single source of truth for data.

⑤. Structured DDB: Organized, formatted, and easily searchable (think tables, rows, columns). Examples: customer info in a CRM system (name, address, phone number).

- Semi-structured DDB: has some organization, but doesn't conform to rigid structure (think XML files, JSON, CSV).
- Unstructured DDB: No predefined structure, hard to search and analyze (think text, images, videos). Examples: customer feedback emails or social media posts.

⑥ DDB modeling is like creating a blueprint or map of your data. It defines how data is structured, stored and controlled, helping it easier for designing systems that work with data.

Two type of DDB models:

* Conceptual DDB model: high-level focusing on entities and relationships (like a business's key data elements).

* Logical DDB Model: More detailed, defining attributes, keys and relationships (like a blueprint for database design).

⑦ Data mining is like digging for gold in a massive data mountain. It's the process of discovering patterns, trends, and insights from large datasets using techniques like machine learning and statistics.

Businesses use data mining to:

- * uncover hidden patterns in customer behavior.
- * predict future trends.
- * optimize operations.

Two real-world examples:

- * Retail: Walmart uses data mining to analyze customer purchase patterns, optimizing inventory and product placement.
- * Finance: Banks use data mining for fraud detection, spotting unusual transaction patterns to prevent unauthorized activity.

⑧ A database management system is like a super-powered tool that helps you store, manage and retrieve data efficiently.

Here are four popular DBMS platforms

- * MySQL: open-source relational DBMS, great for structured data. Use case: web apps like WordPress or e-commerce sites (e.g. Amazon).
- * PostgreSQL: open-source relational DBMS with advanced features. Use case: Geospatial apps (e.g. mapping services) or complex querying needs.
- * Oracle: Commercial relational DBMS with robust enterprise features. Use case: large corporations needing high security and scalability (e.g. banking system).
- * MongoDB: NoSQL document based DBMS, flexible schema. Use case: Handling unstructured data (e.g. social media content, IoT data).

⑨ ETL (Extract Transform Load) is like a data pipeline.

- * Extract: pull data from various sources (databases, files)
- * Transform: clean, format, manipulate data (handle missing values, convert types, etc.)
- * Load: put transformed data into target system (data warehouse, database).

ETL is crucial in data engineering because it:

- * Ensures data quality and consistency.
- * Centralizes data for easier access.

⑩ Common data storage formats used in analytics:

- * CSV (Comma-separated values): Simple, human-readable text format. Use for: small datasets, simple exchanges.
- * JSON (JavaScript Object Notation): Flexible nested data format. Use for: big data, API design, web data (like HTTP).
- * Avro: Compact, schema-based binary format. Use for: serializing data in big data pipelines (like Hadoop).

you'd pick one based on:

- * Readability vs efficiency (CSV is readable, Parquet is efficient)
- * Structure (JSON for flexible, parquet for columns, only + read)

- ⑩ Artificial Intelligence is like teaching machines to think and learn like humans. It's about creating systems that can perform tasks that typically require human intelligence, like understanding language, recognizing images or making decisions.

Differences between Narrow AI and General AI :

* Narrow AI : focuses on specific tasks like playing chess, recommending products). Example: Netflix.

* General AI : A hypothetical AI that can handle any intellectual tasks & humans can like understanding complex contexts, adapting to new situations).

Example : A sci-fi robot that can learn, code and chat like a human.

- ⑪ Machine Learning is like teaching a computer to learn from data instead of being explicitly programmed.

Three main types of machine learning :

* Supervised learning : Trained on labeled data like images, emails, etc. Example : spam email filters.

* Unsupervised learning : Finds patterns in unlabeled data (like grouping similar customers). Example : Market segmentation.

* Reinforcement learning : learns by trial and error, getting rewards for good actions. Example : training a robot to walk, play games or drive.

- (13) Deep learning is a subset of machine learning, which itself is part of artificial intelligence. It's like super-powered ML using neural networks with multiple layers to learn complex patterns in data.

Neural Networks: inspired by the brain's neurons, they're layers of interconnected nodes processing data. Here's how they work:

- Input layer gets data
- Hidden layers extract features (like edges in images)
- Output layer gives predictions

- (14) Natural Language Processing (NLP) is like teaching computers to understand and generate human language. It's a hybrid combining AI, linguistics and computer science to handle text, speech and more.

Three - real world NLP apps you likely use:

- * Google Translate: translates languages in real-time
- * Siri & Google Assistant: understand voice commands and respond.
- * Spam filters in Gmail: classifies emails as spam or not.

- 15) A large language model (LLM) is like a super smart AI trained on massive text data to understand and generate human-like language.

Two well known LLMs:

- * GPT (Generative Pre-trained Transformer): used for text generation, summarization and answering questions.
- * BERT (Bidirectional Encoder Representations from Transformers): used for understanding context in search queries, sentiment analysis and more.

- 16) Generative AI is like an AI that creates new stuff - text, images, music or even code. At a high level it works by learning patterns in training data and generating similar, but new outputs.

Three tools using generative AI:

- Chat GPT: Generates human-like text responses.
- Dall-E: Creates images or videos from text prompts.
- GitHub Copilot: Suggests code snippets based on context.

- (17) A training dataset is like the fuel for an AI model - it is the data used to teach the model patterns and make predictions.

Quality of training data is crucial because:

- * Accuracy: Garbage in garbage out - if data's wrong the model's predictions will be off.
- * Bias: If data's biased, the model learns those biases (like unfair stereotypes).

If training data's biased or incorrect:

- Biased decisions: AI might discriminate (e.g. biased hiring tools).
- Inaccurate predictions: Like a medical AI misdiagnosing based on skewed data.
- Trust issues: Users lose faith in AI systems.

- (18) Computer vision is like giving machines the power to see and understand images or videos. Machines process images using algorithms (like deep learning models) to detect patterns, objects or text.

Two real world use cases:

- * Self driving cars: detecting pedestrians, lanes and obstacles.
- * Face unlock on phones: Recognizing faces for security.

① An API (Application Programming Interface) is like a waiter. It's responsible for you (the customer) order food (request data & service) the waiter takes it to the kitchen (system), and brings back your food (response).

APIs are common in modern software because they:

- Enable systems to talk to each other
- Allow modular, reusable app development
- Let you integrate external services (like payment gateway)

② An API key is like a secret password for accessing an API. Its purpose is to authenticate and authorize access to specific service or data.

Purpose: identify you to the API, track usage and secure access.

How to store and protect it:

- * Use environment variables (not hard coded in code)
- * Store in secure vaults like AWS Secrets Manager?
- * Rotate keys regularly

If exposed publicly

- * misuse: someone could use your quota, hit rate limits or access restricted data.
- * costs: unexpected charges if the API is paid.
- * security risks: leaked data or actions on your behalf.

(21) Rest API is like a single standard way for system to talk over the web. REST stands for Representational State Transfer.

Using HTTP method in REST API:

- * GET: Fetch data (like reading a post).
- * POST: Create new data (like submitting a form).
- * PUT: Update existing data (like editing a profile).
- * DELETE: Remove data (like deleting a file).

(22) JSON is like a light weight, easy to read format for sharing data. It's commonly used with APIs and data exchange because:

- humans can read it.
- machines can parse it easily.
- work with most programming languages.

Single example of JSON

< > json code
view code

```
{  
  "name": "John",  
  "age": 30,  
  "city": "Cape Town"  
}
```

(23) Prompt engineering is like crafting the perfect question to get what you need from AI tools like ChatGPT. It's crucial because:

- AI's output depends on the input
- Good prompts = better results, less hassle

Examples:

- Poor: "write something about South Africa"
- Improved: "Rewrite this sentence in correct English: I go store yesterday".

(24) The map function is like a factory conveyor belt - it applies a function to each item in a list and returns a new list with results.

Example in JupyterScript:

<> JupyterScript code
view code

```
const numbers = [1, 2, 3];  
const doubled = numbers.map(x =>  
  console.log(doubled); // [2, 4, 6] in Python:
```

<> Python code
view code

```
numbers = [1, 2, 3]  
doubled = list(map(lambda x: x * 2,  
  print(doubled) # [2, 4, 6]
```


(25) Cloud computing is like renting computing power - storage, servers, apps - over the internet.

Three main cloud service models:

- IaaS (Infrastructure as a service): Rent servers, storage, networking, example: AWS EC2.
- PaaS (Platform as a service): Platform for developing apps no server management, example: Heroku.
- SaaS (Software as a service): Ready to use apps over the web, example: Gmail.

(26) Version control is like a time machine for your code - tracks changes, lets you revert and collaborate.

Git is a popular version control system.

Dev's inhbits and dev's use it because:

- tracks changes to code/files Enables collaboration (like team projects)
- Easy to revert mistakes

Key Git things

- Commit: Save what I changes with a message.
- Branch: Separate line of development (like working on a feature).

(27) Jupyter notebook is like an interactive coding playground - web-based tool for writing code, visualizing data and sharing notebooks.

mostly used with Python, but supports other languages too.

Popular among data scientists because:

- interactive: Run code cells, see output immediately
- visualizations: Great for plots, charts
- Docs and sharing: mix code, text and results in one doc.

(28) Python is like a Swiss Army knife. - versatile, easy to use programming language.

Popular for data analysis and AI because:

- simple syntax
- massive libraries for data/AI
- Great community.

Here are Python libraries for data work:

- Pandas: for data manipulation and analysis (like Excel on steroids)
- NumPy: for numerical computations (fast math ops).
- Matplotlib: for data visualization (making plots).
- Scikit-learn: for machine learning (building models).

(29) Data privacy is about protecting personal information from being misused or stored without consent. It exists because it safeguards individual's rights, protects identity theft and maintains trust in digital services.

Key aspects of GDPR

- Personal data protection: covers information like names, emails, addresses, ID numbers and online identifiers.
- Individual rights: includes access, correction, erasure and restriction of data processing.
- Organizational accountability: requires data protection measures and breach notifications.

GDPR applies to EU member states (e.g. Germany, France, Italy) and EEA countries (Iceland, Liechtenstein, Norway). The UK has its own version, the UK GDPR, post Brexit.

- (30) Data ethics is like the "do's" and don't's of handling data - ensuring it's collected, used and stored responsibly.

Three examples of unethical data / AI use:

- Example 1: A health app secretly shares users' mental health data with insurance companies. Problematic: violates privacy, could affect insurance premiums.
- Example 2: An AI hiring tool trained on biased past hiring data rejects qualified candidates from underrepresented groups. Problematic: perpetuates discrimination, unfair hiring.
- Example 3: A social media platform manipulates news feeds to increase engagement without consent. Problematic: Exploits users' psychology, harms mental health.

3.1) Data governance is the process of managing the availability, usability, integrity and security of an organization's data.

A good data governance framework has these key components:

1. Data strategy: Aligns data governance with business objectives
2. Data policies: Define rules and standards for data management
3. Data Quality: Ensures data accuracy, completeness and consistency.
4. Data security: Protects sensitive data from unauthorized access.
5. Data Stewardship: Assigns roles and responsibilities for data management.

3.2) Bias in AI is like a skewed lens - the model favors certain outcomes or groups over others often unfairly.

An AI model is biased when it:

- Reflects training data's prejudices
- Makes unfair predictions for some groups.

Real-World Example: Facial recognition AI struggles with darker skin tones (like fewer accurate matches for Black faces).
• consequence: wrong identifications could lead to misidentifications could lead to misidentification by police, unfair arrests or denied services.

(33) Role	Focus	key skills	common tools
DATA Analyst	Analyze data Create reports	SQL, Excel, Stats	Tableau, Power BI, SQL
DATA Scientist	Build models, Predict Outcome	Python, R, ML	Jupyter, Scikit-learn, TensorFlow
DATA Engineer	Build data Pipelines, Infrastructure	SQL, Python, Cloud	Apache Spark, AWS, Google Cloud
AI Engineer	Deploy AI Models, build	Python, ML, Software dev	TensorFlow, PyTorch

(34) Business intelligence (BI) is like a business eyes - turning data into insights for better decisions.

Two popular BI tools:

- Power BI: creates dashboards, reports from data (like a customizable business scorecard).
- Tableau: visualizes data, finds patterns like turning numbers into stories.

How they help:

- Spot trends: see what's selling.
- Track performance: monitor goals vs. reality.
- Make predictions: forecast based on data.
- Share insights: collaborate across teams.

(25) The concept of bias in AI caught my attention.

It's crazy how model can perpetuate all world inequalities like facial recognition struggles with darker skin tones.

The concept of bias in AI is like a mirror reflecting societal prejudices. When models are trained on skewed data, they amplify existing inequalities.

This can lead to unfair outcomes perpetuating stereotypes and limiting opportunities. It is vital to address these biases to ensure AI systems are fair and inclusive.

(26) Data ethics and AI bias I find them very complex topics

- Data ethics : I think of it as digital responsibility - ensuring data is collected, used and shared fairly.

AI bias : AI systems can reflect and amplify human biases, leading to unfair outcomes. Which makes it more complex to easily understand.

Some helpful resources

- Books : "Weapons of Math Destruction" by Cathy O'Neil.
- Online courses : Coursera's AI ethics courses.
- Articles : Brookings Institution on AI bias.

(37) Data and AI are like new electricity
- They'll power many industries. In South Africa
we see big impact in:

- Healthcare: AI predicting disease outbreaks & personalizing treatments.
- Finance: Smarter fraud detection, personalized financial advice.
- Transport: AI optimizing traffic, maybe even self-driving taxis.

Everyday life could see:

- Personal Assistance: AI helps managing schedules, like Google Assistant.
- Smart Homes: AI optimizing energy use, security.
- Shopping: AI recommending products, maybe even cashier-less stores.